New chat

L9_S1_A (1).pdf
PDF

Example Lecture Scribe (Structure Only)

Definitions and Notation
Present the formal definitions and symbols introduced in the lecture.

Assumptions
List all assumptions explicitly stated during the lecture.

Theorems / Results
State all theorems and results exactly as covered in the lecture material.

Proofs / Derivations
Provide step-by-step derivations strictly following what is shown in the lecture.

Worked Examples
Include solved examples with intermediate steps only where they appear in the lecture.

Conclusion
Mention only the key outcomes discussed in the lecture.

You are provided with pdf of lecture material. Your task is to generate a lecture scribe suitable for exam reference, strictly following the structure above.

Use only the content from the provided slides and relevant textbook sections.

Include definitions, notation, assumptions, results, proofs, derivations, and examples only if they are explicitly taught in the lecture.

Show steps only up to the point demonstrated in the

lecture.

Maintain a formal academic tone and follow the lecture flow precisely.

Do not add explanations, intuition, summaries, new theory, or new examples.

Do not fill in missing steps.

Ignore informal content and images.

Do not infer or introduce any information that is not provided.

Below is the lecture scribe prepared strictly from the provided slides, organized exactly in your requested structure and limited to material explicitly shown in Lecture 9. No additional theory or steps have been introduced.

Lecture 9 Scribe — Uniform, Exponential, Laplace, and Gamma Random Variables

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Definitions and Notation

Continuous Random Variables — Uniform

Let $X \sim \text{Uniform}(a, b)$, $a < b$.

PDF

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{elsewhere} \end{cases}$$

CDF

$$F_X(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a < x < b \\ 1, & x > b \end{cases}$$

Exponential Random Variable

For parameter $b > 0$:

PDF

$$f_X(x) = \frac{1}{b} e^{-x/b} u(x)$$

CDF

$$F_X(x) = [1 - e^{-x/b}]\, u(x)$$

Laplace Random Variable

PDF

$$f_X(x) = \frac{1}{2b} \exp(-\frac{|\, x \,|}{b})$$

CDF

$$F_X(x) = \begin{cases} \frac{1}{2} e^{x/b}, & x < 0 \\ 1 - \frac{1}{2} e^{-x/b}, & x > 0 \end{cases}$$

Gamma Random Variable

PDF

$$f_X(x) = \frac{(x/b)^{c-1} e^{-x/b}}{b\, \Gamma(c)} u(x)$$

CDF

$$F_X(x) = \frac{\gamma(c, x/b)}{\Gamma(c)}$$

Gamma Function

$$\Gamma(a) = \int_0^\infty t^{a-1} e^{-t}\, dt$$

Incomplete Gamma Function

$$\gamma(a, x) = \int_0^x t^{a-1} e^{-t}\, dt$$

Assumptions

- Uniform RV assumes all values in $[a, b]$ are equally likely.
- Exponential RV is defined for $x \geqslant 0$.
- Laplace RV is symmetric about zero.
- Gamma RV parameters are positive.
- Unit step function $u(x)$ is used to enforce support.
- In optimization example, travel time is a continuous RV with PDF $f(t)$ and CDF $F(t)$.

(All stated directly in lecture slides.)

Theorems / Results

Special Cases of Gamma RV

- For $b = 2,\ c = 0.5$: Chi-square random variable.
- For $c = 1$: Exponential random variable.

Movie Theater Optimization Result

Optimal departure time:

$$d^* = T - F^{-1}\left(\frac{k}{c+k}\right)$$

Interpretation (as stated): depart such that probability of arriving early equals $\frac{k}{c+k}$.

Proofs / Derivations

Example 2 — Transformation of Exponential RV

Given:

$$f_X(x) = e^{-x}u(x), \quad F_X(x) = [1 - e^{-x}]\,u(x)$$

(a)

$$\Pr(3X < 5) = \Pr(X < \frac{5}{3}) = F_X(\frac{5}{3}) = 1 - e^{-5/3}$$

(b)

$$\Pr(3X < y) = F_X(\frac{y}{3}) = [1 - e^{-y/3}]\,u(y)$$

(c) Let $Y = 3X$.
CDF obtained from (b). Differentiation gives $f_Y(y)$.
Conclusion: Y is exponential with parameter $\lambda = 1$.

Example 4 — Gamma Mean and Variance

Given $X \sim \text{Gamma}(a, \lambda)$:

Mean

$$E[X] = \frac{a}{\lambda}$$

(using textbook Eq. 6.1 as cited in slides)

Variance

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

Intermediate steps shown:

$$E[X^2] = \frac{\Gamma(a+2)}{\lambda^2 \Gamma(a)}$$

Final result:

$$\mathrm{Var}(X) = \frac{a}{\lambda^2}$$

Worked Examples

Example 1 — Uniform RV (Phase)

Given:

$$f_\Theta(\theta) = \frac{1}{2\pi}, \quad 0 < \theta < 2\pi$$

(a)

$$\Pr(\Theta > 3\pi/2) = \frac{1}{4}$$

(b)

$$\Pr(\Theta < \pi \mid \Theta > 3\pi/2) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

(c)
Using cosine inequality and interval mapping:

$$\Pr(\cos\Theta < 1/2) = \frac{4\pi/3 - 2\pi/3}{2\pi}$$

Example 3 — Laplace RV

Given:

$$f_W(w) = ce^{-2|w|}$$

(a) Normalization:

$$\int_{-\infty}^{\infty} ce^{-2|w|}\,dw = 1 \Rightarrow c = \frac{1}{4}$$

(b)

$$\Pr(-1 < W < 2)$$

(c)

$$\Pr(W > 0 \mid -1 < W < 2) = \frac{\Pr(0 < W < 2)}{\Pr(-1 < W < 2)}$$

with intermediate integrals shown explicitly on slides.

Problem Solving 2 — Given CDF

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 < x < 1 \\ 1, & x > 1 \end{cases}$$

PDF:

$$f_X(x) = 2x,\ 0 < x < 1$$

Results:

- Mean: $\mu_X = \frac{2}{3}$
- Variance: $\sigma_X^2 = \frac{1}{18}$
- Skewness: negative
- Kurtosis: platykurtic

Conclusion

- Formal PDFs and CDFs for Uniform, Exponential, Laplace, and Gamma RVs were introduced.
- Special Gamma cases yield Exponential and Chi-square distributions.

- Transformation of RVs was demonstrated via scaling.
- Mean and variance of Gamma RV were derived.
- Conditional probability and optimization using CDFs were applied.
- Final results included optimal departure time and statistical moments from a given CDF.

If you'd like, next I can convert this into a printable exam-reference PDF or LaTeX format.